RETAIL BILLING SYSTEM

Python Programming Case Study

1. Problem Statement

Preparing a retail bill manually for several products can take time and may lead to calculation errors. The bill must include each product's price and quantity, category-based tax, and any discount based on the total purchase amount. A Python program can automate these calculations and display the final bill.

2. Objective

To create a Python-based retail billing system that accepts product details, calculates item amounts, applies different tax rates for food and other products, calculates eligible discounts with a maximum discount limit, and prints the total amount, discount, and final bill.

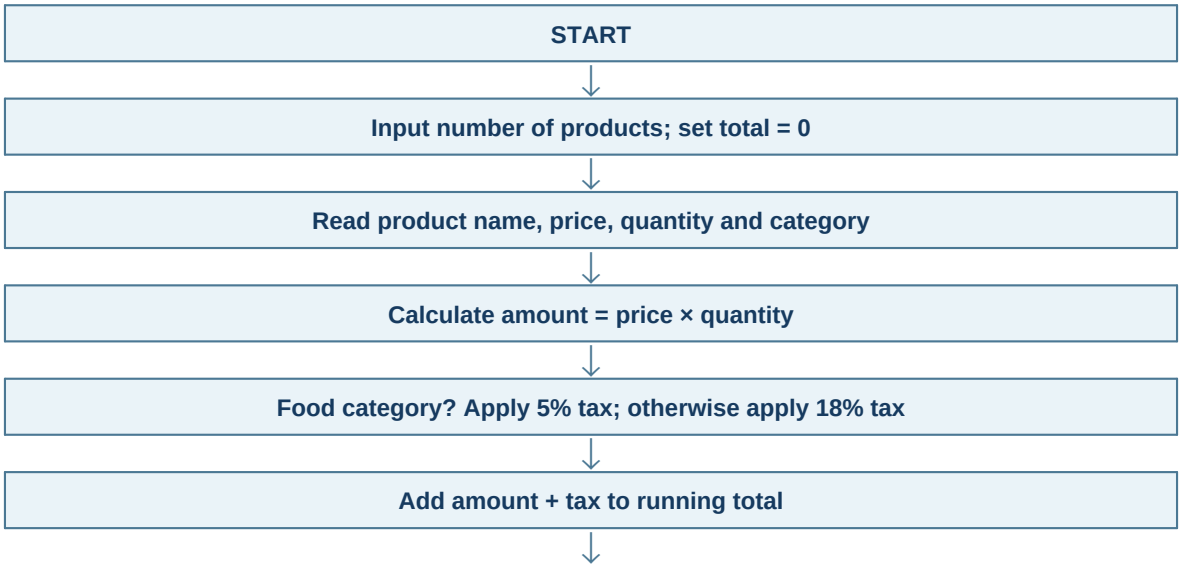
3. Assumptions

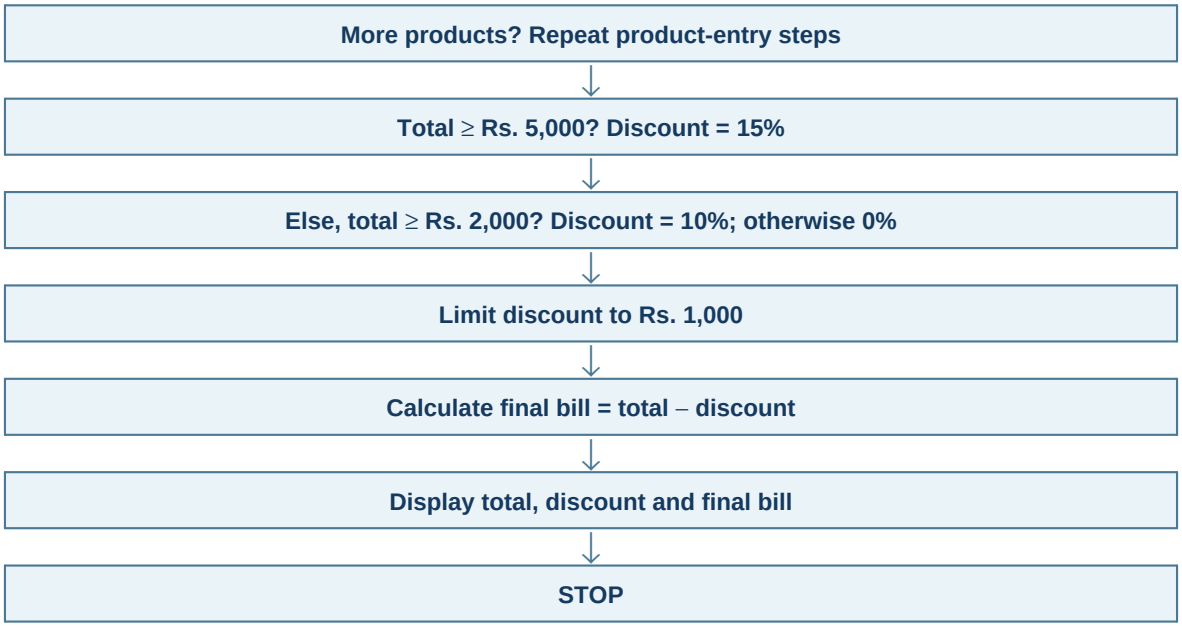
- The user enters a valid number of products, product names, prices, quantities, and categories.
- Product price and quantity are non-negative values, and category is entered as food or other.
- Food products are taxed at 5%; other products are taxed at 18%.
- A 15% discount applies when the running total is at least Rs. 5,000.
- A 10% discount applies when the running total is at least Rs. 2,000 but below Rs. 5,000.
- No discount applies when the total is below Rs. 2,000.
- The maximum discount is limited to Rs. 1,000.

4. Algorithm

1. Start the program and read the number of products.
2. Initialize the total to zero.
3. For each product, read its name, price, quantity, and category.
4. Calculate the amount as price multiplied by quantity.
5. If the category is food, calculate tax at 5%; otherwise calculate tax at 18%.
6. Add the amount and tax to the running total, then display the product details.
7. After all products are processed, calculate the discount: 15% for totals of Rs. 5,000 or more, 10% for totals of Rs. 2,000 or more, otherwise zero.
8. Limit the discount to a maximum of Rs. 1,000.
9. Subtract the discount from the total to calculate the final bill.
10. Display the total amount, discount, and final bill. Stop.

5. Flowchart



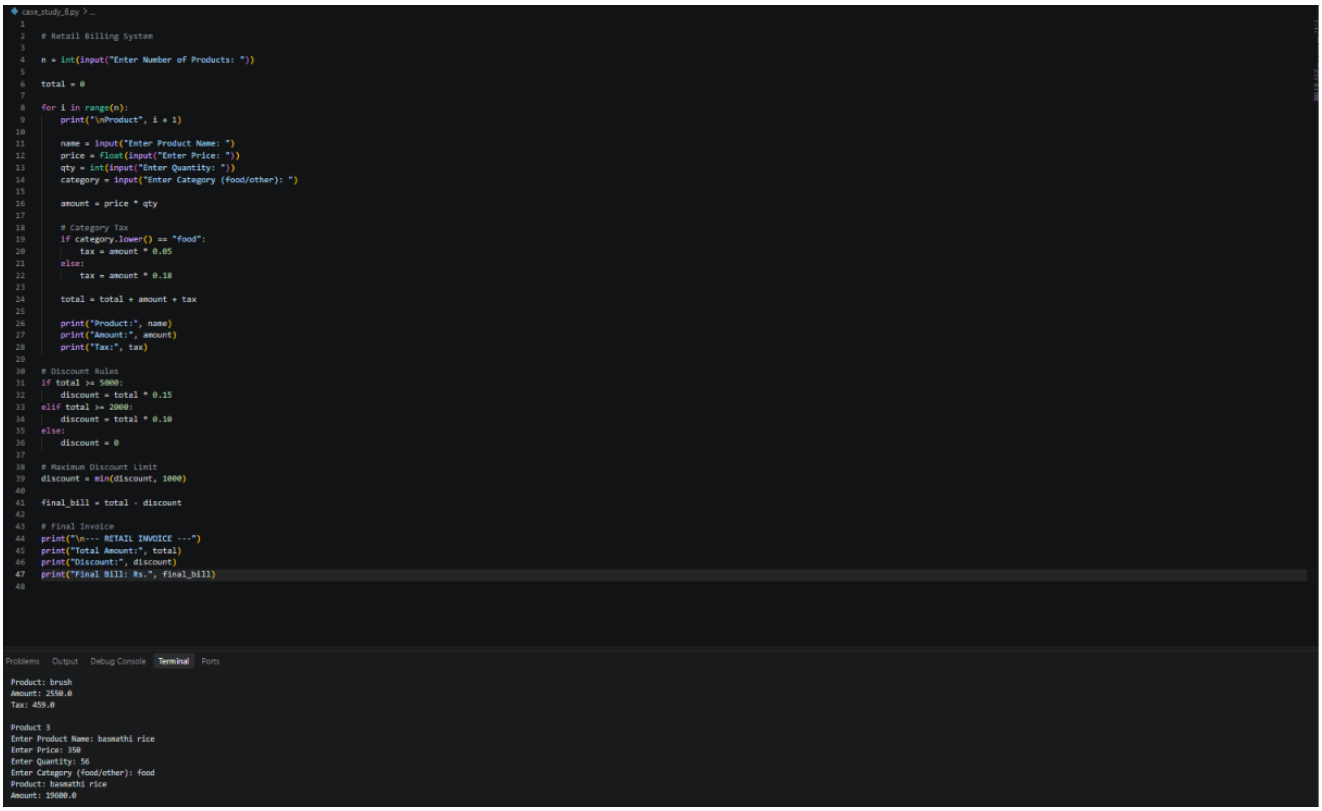

More products? Repeat product-entry steps
Total ≥ Rs. 5,000? Discount = 15%
Else, total ≥ Rs. 2,000? Discount = 10%; otherwise 0%
Limit discount to Rs. 1,000
Calculate final bill = total − discount
Display total, discount and final bill
STOP

6. Pseudocode

```
BEGIN
INPUT number_of_products
total = 0
FOR each product
INPUT name, price, quantity, category
amount = price * quantity
IF category is food THEN
tax = amount * 0.05
ELSE
tax = amount * 0.18
END IF
total = total + amount + tax
DISPLAY name, amount, tax
END FOR
IF total >= 5000 THEN discount = total * 0.15
ELSE IF total >= 2000 THEN discount = total * 0.10
ELSE discount = 0
END IF
discount = MIN(discount, 1000)
final_bill = total - discount
DISPLAY total, discount, final_bill
END
```

7. Python Source Code and Output Screenshot

The original screenshot supplied for this case study is included on the next page.



```
case_study_6.py > ...
1
2   # Retail Billing System
3
4   n = int(input("Enter Number of Products: "))
5
6   total = 0
7
8   for i in range(n):
9       print("\nProduct", i + 1)
10
11      name = input("Enter Product Name: ")
12      price = float(input("Enter Price: "))
13      qty = int(input("Enter Quantity: "))
14      category = input("Enter Category (food/other): ")
15
16      amount = price * qty
17
18      # Category Tax
19      if category.lower() == "food":
20          tax = amount * 0.05
21      else:
22          tax = amount * 0.18
23
24      total = total + amount + tax
25
26      print("Product:", name)
27      print("Amount:", amount)
28      print("Tax:", tax)
29
30  # Discount Rules
31  if total >= 5000:
32      discount = total * 0.15
33  elif total >= 2000:
34      discount = total * 0.10
35  else:
36      discount = 0
37
38  # Maximum Discount Limit
39  discount = min(discount, 1000)
40
41  final_bill = total - discount
42
43  # Final Invoice
44  print("\n--- RETAIL INVOICE ---")
45  print("Total Amount:", total)
46  print("Discount:", discount)
47  print("Final Bill: Rs.", final_bill)
48

Problems  Output  Debug Console  Terminal  Ports

Product: brush
Amount: 2550.0
Tax: 459.0

Product 3
Enter Product Name: basmathi rice
Enter Price: 350
Enter Quantity: 56
Enter Category (food/other): food
Product: basmathi rice
Amount: 19600.0
```

8. Conclusion

The Retail Billing System automates product-wise amount and tax calculations, applies discounts according to purchase totals, and displays the final bill. It demonstrates the practical use of Python input, loops, conditional statements, arithmetic operations, and built-in functions such as min().